

**SafeRoute+: A Cloud and GIS-Based Predictive System
for Evacuation Center Oversight**

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CHAPTER I

INTRODUCTION

Project Description and Outline

SafeRoute+ is a cloud and GIS-based predictive system designed to support the monitoring and management of evacuation centers during disasters. The system utilizes spatial data, real-time updates, and analytical tools to enable local officials to identify available shelters, check capacity levels, plan safe routes, and make informed decisions during emergencies. The project aims to create a simple and accessible platform that can guide communities and authorities before and during evacuation activities.

The system has three main functions. First is the GIS mapping feature for locating evacuation centers and viewing hazard-prone areas. Second is the cloud-based database for storing and updating information on shelter capacity, accessibility, and safety conditions. Third is the predictive module that estimates potential congestion and suggests route options. By combining these features, SafeRoute+ supports disaster response through organized data, clear visualization, and timely decision-making.

The chapter presents an overview of the project, including its purpose, objectives, conceptual direction, significance, scope, limitations, and the key terms used in the study.

Objectives of the Study

The general objective of this study is to design and develop SafeRoute+: A Cloud and GIS-Based Predictive System for Evacuation Center Oversight.

Specifically, it aims to:

1. Design and develop a web-based platform capable of mapping evacuation centers using GIS tools.
2. Integrate cloud storage for the secure and accessible handling of evacuation related data.
3. Implement predictive analytics that estimate shelter availability and potential congestion during emergencies.
4. Provide real time monitoring features that help officials make quick and accurate decisions.

Conceptual Framework

The conceptual framework of SafeRoute+ is anchored on three core components which work together to support evacuation management.

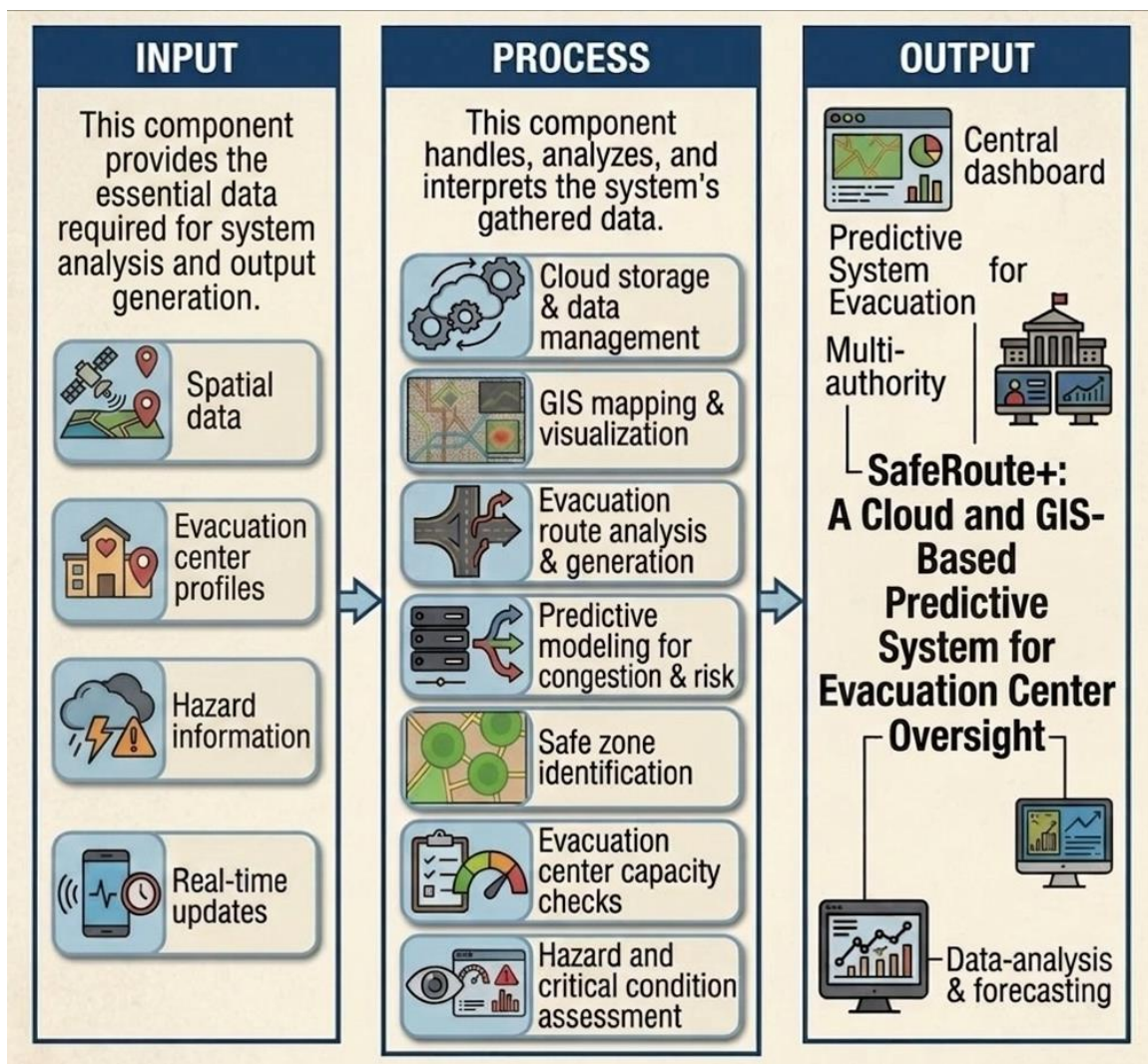


Figure 1. Conceptual Framework

Significance of the Project

SafeRoute+ offers benefits to various groups involved in disaster preparedness and evacuation management. The system aims to help the following:

Local Government Units. The system provides real-time information on evacuation centers and hazard conditions. This supports faster and more accurate decisions during emergencies and helps in resource allocation.

Disaster Response Teams. GIS-based route guidance and predictive alerts assist responders in planning safe evacuation paths and avoiding risk areas. Updated information on shelter status also improves coordination.

Communities. Residents gain improved access to timely and reliable evacuation guidance. With clearer information, evacuation becomes more organized, reducing confusion and potential casualties.

Researchers and Planners. The platform helps showcase how cloud computing and GIS technologies can be applied to local disaster challenges, offering a reference for further system development.

Overall, the project promotes community safety and

resilience by enhancing the management of evacuation centers and emergency routes.

Scope and Limitation of the Project

The project will focus on creating a functional prototype of a cloud and GIS-based monitoring system. It includes features such as interactive maps, evacuation center profiles, route visualization, and simple predictive alerts. The geographic coverage will be limited to selected barangays or areas chosen for testing. System development will concentrate on web-based access for practicality and ease of deployment.

The system will not function as a full-scale operational emergency platform. Real-time hazard data, such as rainfall measurements, flood depth, or sensor readings, will rely on already available datasets and will not include the installation of physical sensors. Predictive outputs are limited to estimated trends and do not guarantee exact forecasts. The accuracy of the system will depend on the completeness and quality of the data provided to it.

Definition of Terms

To ensure clarity, the following terms are defined as used in the project:

Cloud Computing

A technology that allows storage, processing, and access to data through online servers instead of local machines.

Geographic Information System or GIS

A tool used for mapping, analyzing, and displaying spatial data related to locations, routes, and environmental conditions.

Predictive Analytics

A method of using existing data to estimate possible outcomes, such as shelter congestion or route risk.

Evacuation Center

A facility designated for the temporary shelter of individuals during disasters or emergencies.

Evacuation Route

A path recommended for safe movement of people from hazard areas to designated evacuation centers.

Real Time Monitoring

The ability of the system to update information instantly as new data becomes available.

Spatial Data

Information that refers to physical locations or geographic features on a map.

CHAPTER II

REVIEW OF RELATED LITERATURE/SYSTEM

Effective disaster management requires accurate data, rapid decision-making, and coordinated response systems. The Philippines faces compounded challenges due to its exposure to multiple natural hazards, demanding systematic risk assessment and evacuation planning that account for local topography, infrastructure, and population density (Andrada et al., 2020; Arao, 2020; Gacu et al., 2022). In response, researchers have increasingly integrated advanced technologies into disaster operations. Cloud computing offers scalable, cost-effective emergency response solutions with real-time data processing capabilities critical for time-sensitive situations (Ahmed et al., 2020; Ayoub et al., 2020; Hong et al., 2021). Geographic Information Systems have become essential for spatial analysis, enabling flood risk assessment, shelter identification, and evacuation route planning through multicriteria decision analysis (Chelariu et al., 2022; Parajuli et al., 2023; Yao et al., 2021). Predictive analytics and machine learning have further transformed preparedness by enabling proactive risk assessment and dynamic response planning, with AI-powered systems demonstrating high prediction accuracy across

multiple data sources (Chen et al., 2020; Ermino et al., 2022; Ghorbanzadeh et al., 2018). This review synthesizes local and foreign literature thematically across technological, analytical, and implementation dimensions of these tools to identify relevant practices and research gaps informing the development of SafeRoute+: A Cloud and GIS-Based Predictive System for Evacuation Center Oversight.

REVIEW OF RELATED LITERATURE AND SYSTEM

Disaster Risk Assessment and Evacuation Planning

Studies on disaster risk assessment and evacuation planning consistently emphasize the need for systematic approaches that account for local geographic conditions, infrastructure limitations, and population density. In the Philippine context, researchers such as Andrada et al. (2020), Arao (2020), and Gacu et al. (2022) found that many municipalities, particularly those prone to flooding and earthquakes, require reliable GIS-based decision support systems capable of reflecting local topography, infrastructure conditions, and population density. Their findings reveal how risk mapping efforts can identify highly susceptible zones, with some studies noting flood vulnerability in up to 56 percent of assessed areas. These

works also point out variabilities in local government preparedness, including uneven implementation of alert systems, limited access to training, and challenges in resource distribution (Dey et al., 2024). The implementation of disaster preparedness programs in Philippine cities has shown varying levels of effectiveness, with significant gaps identified in alert system registration, training access, and resource allocation across different municipalities (Andrada et al., 2020; Arao, 2020). Furthermore, Gacu et al. (2022) and Torno (2024) illustrated that local government units in the Philippines often face challenges in implementing comprehensive disaster risk reduction measures due to limited technical capacity, inadequate funding, and insufficient interagency coordination.

Internationally, a substantial body of research highlights the value of GIS in emergency planning. Chelariu et al. (2022), Parajuli et al. (2023), and Yao et al. (2021) demonstrate how GIS tools support detailed hazard assessments, shelter planning, and evacuation route optimization. These approaches often use analytical models such as Analytical Hierarchy Process and network analysis to assess suitability based on accessibility, capacity, and environmental risk (Gacu et al., 2022; Otun & Arekhandia,

2020; Shkedova et al., 2024). The integration of GIS with various analytical techniques has proven effective in evaluating evacuation center suitability based on multiple parameters such as accessibility, capacity, and safety (Gacu et al., 2022; Otun & Arekhandia, 2020; Shkedova et al., 2024). Similarly, studies from Aydin and Birincioğlu (2022), Edirisinghe et al. (2021), and Ghorbanzadeh et al. (2018) emphasize the importance of integrating real-time environmental data to update hazard maps and improve decision-making under rapidly changing conditions. GIS-based flood hazard mapping, combined with real-time data integration, significantly enhances decision-making capabilities during emergency situations, particularly in identifying high-risk areas and optimizing resource allocation (Aydin & Birincioğlu, 2022; Edirisinghe et al., 2021; Ghorbanzadeh et al., 2018).

Community-centered evacuation planning emerges as a critical theme across both local and international literature. Campolet et al. (2025), Arao (2020), and Torno (2024) highlighted that evacuation initiatives in Philippine communities benefit significantly from combining local knowledge with technical methods such as GIS-based mapping. Community-driven programs were shown to improve disaster

awareness, with some initiatives reporting up to 82 percent increases in plan familiarity among residents. The establishment of barangay-level disaster risk reduction and management committees, combined with GIS-based evacuation planning tools, has proven effective in improving local disaster preparedness and response capabilities (Andrada et al., 2020; Campolet et al., 2025). These studies stress the need to account for urban congestion, cultural factors, rural terrain conditions, and transportation limitations when planning localized evacuation strategies. Similarly, Afolabi et al. (2023) and Torno (2024) noted that effective systems require flexible structures that adapt to varying levels of hazard awareness and infrastructure capacity across barangays and municipalities, emphasizing that Philippine evacuation systems must address specific local challenges including limited transportation infrastructure, high population density in urban areas, and varying levels of disaster awareness among different community segments.

On an international scale, Afolabi et al. (2023), Cvetković (2024), and Torno (2024) demonstrate that community involvement strengthens disaster literacy, improves risk communication, and increases evacuation compliance. Community-based disaster risk reduction systems, when

integrated with digital tools and GIS technologies, significantly enhance community resilience by empowering local stakeholders with real-time risk information and participatory planning tools, resulting in improved emergency response coordination and reduced casualties (Afolabi et al., 2023; Cvetković, 2024; Torno, 2024). Successful community-based disaster risk reduction implementation requires integration of local knowledge with scientific approaches, continuous training programs, and establishment of clear communication channels between authorities and community members (Andrada et al., 2020; Arao, 2020; Makoto et al., 2024).

Cloud Computing, GIS, and Technological Infrastructure

Cloud-based technologies have become instrumental in supporting largescale disaster operations due to their scalability, reliability, and ability to process vast datasets in real time. Studies by Ahmed et al. (2020), Ayoub et al. (2020), and Hong et al. (2021) show how cloud infrastructures outperform traditional systems by enabling continuous access, faster computation, and automated resource distribution. These capabilities are essential for maintaining system availability during emergencies. The integration of cloud technologies enables disaster management

systems to handle massive amounts of data while maintaining high availability and reliability, as evidenced by the development of various cloud-based emergency management platforms (Ahmed et al., 2020; Gangal, 2025; Hong et al., 2021; Nur et al., 2018). Additionally, Yan et al. (2019) and Leite et al. (2018) provide evidence that real-time data sharing through cloud platforms can shorten emergency response times by as much as 77 percent, demonstrating their critical role in evacuation management. Cloud-based systems significantly improve emergency response coordination through real-time data sharing and automated resource allocation, reducing response times in evacuation scenarios (Yan et al., 2019; Leite et al., 2018).

In the Philippine setting, research on the adoption of advanced technologies shows gradual but promising progress. Studies by Hendriawan Nur et al. (2022), Torno (2024), and Gacu et al. (2022) show that cloud-based and GIS-assisted platforms can significantly improve information sharing, coordination, and resource allocation when implemented within local government units. These digital platforms have already been used to enhance transparency and communication in community-based disaster relief operations, with implementations such as community-based disaster relief

management systems demonstrating improved transparency, coordination, and trust among stakeholders (Torno, 2024; Silagan et al., 2026). However, challenges persist, particularly in smaller municipalities, including limitations in ICT infrastructure, technical training gaps, and financial constraints (Campolet et al., 2025; Gacu et al., 2022). Local literature consistently emphasizes the need for accessible, low-cost technological solutions that align with the capacity of local government units, especially in rural and resource-limited areas. Successful technology adoption in Philippine disaster management requires addressing barriers including limited IT infrastructure, insufficient technical training, and budget constraints, particularly in resource-limited municipalities (Campolet et al., 2025; Gacu et al., 2022).

Real-time data processing and visualization capabilities are crucial for effective emergency response and decision-making during disasters. Hendriawan Nur et al. (2022), Lee et al. (2025), and Yoo and Choi (2019) demonstrated that cloud-based GIS platforms with real-time data processing capabilities can significantly improve emergency response efficiency by providing decisionmakers with up-to-date situational awareness and predictive analytics, reducing evacuation decision times by up to 40 percent. The

implementation of 3D visualization technologies and Web-GIS applications enables intuitive representation of complex spatial data, facilitating better understanding of hazard zones and evacuation routes among both emergency responders and affected populations (Santhanavanich et al., 2020; Shkedova et al., 2024; Yao et al., 2021). The integration of multiple data streams, including satellite imagery, weather forecasts, and social media feeds, through real-time processing frameworks has proven effective in enhancing situational awareness and supporting adaptive emergency management strategies (Chen et al., 2020; GonzálezVilla et al., 2022; Matsuo et al., 2011). Additionally, the application of geospatial analysis in urban planning contexts has demonstrated the value of GIS-based assessments in reflecting user perception and supporting decision-making processes (Stessens et al., 2020; Najafi et al., 2024).

Predictive Analytics, and Evacuation Optimization

Predictive analytics and machine learning have seen increasing application in disaster response across both local and international contexts. Chen et al. (2020), Ghorbanzadeh et al. (2018), and Inoguchi (2024) reported that deep learning models can predict hazard susceptibility with high accuracy, often exceeding 84 percent. The integration of these models

with GIS platforms enables dynamic, real-time assessment, a capability reinforced by studies from Ermino et al. (2022), González-Villa et al. (2022), and Zhu et al. (2019), which show that AI systems can process multisource data such as sensor networks, weather forecasts, and social media activity to improve evacuation decision-making. Machine learning algorithms, can accurately predict disaster susceptibility and optimize evacuation strategies with prediction accuracies exceeding 84 percent (Chen et al., 2020; Ghorbanzadeh et al., 2018; Inoguchi, 2024). The integration of artificial intelligence with GIS technologies enables real-time risk assessment and dynamic evacuation route planning, as shown in studies where AI-powered systems successfully identified urgent disaster-related communications and optimized evacuation routes under various emergency scenarios (Diehr, Ogunyiola, & Dada, 2025; Ermino et al., 2022; González-Villa et al., 2022;). Additionally, machine learning-based systems can process multisource data streams, including sensor networks, social media feeds, and environmental monitoring systems, to provide comprehensive situational awareness and improve evacuation decision-making processes (Pradeep et al., 2024; Santhanavanich et al., 2020; Zhu et al., 2019).

Foreign research on evacuation center management and

route optimization further demonstrates the importance of multifactor analysis. Studies note that effective evacuation planning must account for shelter capacity, distance, population distribution, road conditions, and hazard exposure (Gacu et al., 2022; Matsuo et al., 2011; Yoo & Choi, 2019). Optimal evacuation strategies must consider factors including shelter capacity, accessibility, environmental safety, and real-time population distribution, with studies showing that proper evacuation planning can reduce evacuation times by up to 65 percent (Gacu et al., 2022; Matsuo et al., 2011; Yoo & Choi, 2019). Advanced routing methods such as combined Dijkstra and ant colony optimization have been shown to reduce evacuation times by up to 65 percent (Parajuli et al., 2023; Shkedova et al., 2024). The implementation of network analysis techniques, particularly the integration of Dijkstra algorithm with ant colony optimization, has proven effective in identifying optimal evacuation routes that balance distance minimization with risk avoidance (Gacu et al., 2022; Parajuli et al., 2023; Shkedova et al., 2024). These findings stress the need for adaptable routing systems capable of recalibrating based on traffic, road blockages, and evolving hazard conditions. Successful evacuation systems require dynamic routing capabilities that adapt to changing hazard conditions, traffic congestion, and population mobility

patterns, incorporating factors such as road capacity, flood depth, and pedestrian flow dynamics (Jamrussri & Toda, 2018; Mioc et al., 2008; Simonovic & Ahmad, 2005).

Multi-Criteria Decision Analysis and Hazard Susceptibility Mapping

Foreign studies point to the effectiveness of multicriteria decision analysis frameworks when integrated with GIS. Aydin and Birincioğlu (2022), Jamrussri and Toda (2018), and Ghorbanzadeh et al. (2018) illustrate how MCDA improves evacuation planning and hazard mapping by incorporating environmental and infrastructural factors such as elevation, slope, land use, and hazard proximity. Multicriteria decision analysis approaches, particularly when combined with GIS technologies and Analytical Hierarchy Process methodologies, enable systematic evaluation of multiple risk factors including elevation, slope, land use, and proximity to hazard sources, achieving risk prediction accuracies exceeding 80 percent (Aydin & Birincioğlu, 2022; Ghorbanzadeh et al., 2018; Jamrussri & Toda, 2018; Skilodimou et al., 2019). The application of MCDA frameworks facilitates objective decision-making in shelter site selection and evacuation route planning by integrating diverse criteria such as safety, accessibility, and capacity constraints

(Chelariu et al., 2022; Gacu et al., 2022; Yao et al., 2021). Frequency ratio and weight-of-evidence techniques have also been shown to enhance hazard susceptibility mapping, especially in areas with limited historical data (Dahri & Abida, 2017; Seejata et al., 2018; Wojtek et al., 2019). Frequency ratio methods and weight-of-evidence approaches, when integrated with MCDA, provide robust frameworks for flood susceptibility mapping and emergency response planning, particularly in areas with limited historical disaster data (Dahri & Abida, 2017; Seejata et al., 2018; Wojtek et al., 2019).

Mobile and WebBased Emergency Applications

Mobile and web-based applications have become essential tools for disaster communication and evacuation guidance. Ermino et al. (2022), Inoguchi (2024), and Shitole et al. (2024) demonstrated that mobile applications integrated with real-time hazard monitoring and navigation systems can effectively guide evacuees to safe locations while avoiding high-risk areas, with user studies showing evacuation time reductions compared to traditional evacuation methods. The development of cross-platform emergency applications, incorporating features such as push notifications, real-time traffic updates, and location-based services, has proven

effective in reaching diverse populations during emergencies (Castro et al., 2023; Iqbal et al., 2025; Santhanavanich et al., 2020). Successful mobile emergency applications must balance functionality with usability, ensuring accessibility for users with varying technological literacy while providing comprehensive emergency guidance and real-time updates (Lee et al., 2025; Pradeep et al., 2024; Yan et al., 2019). Applications such as SafeRoute Scanner have demonstrated the potential of comprehensive systems for hazard detection, mapping, and automated reporting to local authorities (Shitole et al., 2024).

Mobile and web-based systems also play a vital role in engaging communities, with applications offering real-time hazard alerts, navigation, and situational updates (Ermino et al., 2022; Castro et al., 2023; Lee et al., 2025). These systems have been shown to reduce evacuation time by up to 30 percent when combined with dynamic mapping and integrated hazard monitoring. The implementation of community-driven evacuation planning, supported by mobile applications and web-based platforms, has shown effectiveness in disaster-prone areas, with studies indicating that community engagement in disaster preparedness activities increases evacuation compliance (Campolet et al., 2025; Inoguchi, 2024;

Mancini et al., 2025; Torno, 2024).

Synthesis

The reviewed literature collectively demonstrates that effective disaster management systems require integrated approaches combining cloud computing, GIS technologies, predictive analytics, real-time visualization, mobile applications, and community engagement. The convergence of these technologies enables comprehensive solutions that address multiple aspects of emergency preparedness, response, and recovery.

Cloud Computing and GIS Integration. Cloud computing emerges as a fundamental enabler for modern disaster management systems, providing scalable infrastructure and real-time data processing capabilities essential for time-critical emergency situations (Ahmed et al., 2020; Hong et al., 2021; Nur et al., 2018). When integrated with GIS technologies, cloud platforms facilitate sophisticated spatial analysis, risk assessment, and evacuation planning across diverse geographic scales (Aydin & Birincioğlu, 2022; Chelariu et al., 2022; Yao et al., 2021). This integration proves particularly valuable in developing countries like the Philippines, where resource constraints and infrastructure limitations necessitate cost-effective, accessible solutions (Gacu et al., 2022;

Hendriawan Nur et al., 2022; Torno, 2024).

Predictive Analytics and Machine Learning. Predictive analytics and machine learning represent crucial advancements in disaster management capabilities, enabling proactive risk assessment and dynamic response planning. Studies demonstrate that AI-powered systems can achieve high prediction accuracies while processing multiple data sources simultaneously, significantly improving situational awareness and decision-making efficiency (Chen et al., 2020; Ermino et al., 2022; Ghorbanzadeh et al., 2018). The application of these technologies to evacuation planning shows particular promise, with optimization algorithms successfully balancing multiple objectives including safety, efficiency, and accessibility (Gacu et al., 2022; Parajuli et al., 2023; Shkedova et al., 2024).

Multi-Criteria Decision Analysis. Multicriteria decision analysis frameworks provide essential methodological foundations for objective disaster risk assessment and evacuation planning. When combined with GIS technologies and supported by comprehensive data sources, MCDA approaches enable systematic evaluation of complex, multidimensional emergency management challenges (Aydin & Birincioğlu, 2022; Ghorbanzadeh et al., 2018; Vojtek et al., 2019). These

frameworks prove particularly valuable in contexts requiring integration of diverse stakeholder perspectives and competing priorities (Jamrussri & Toda, 2018; Seejata et al., 2018). Frequency ratio and weight-of-evidence techniques have been shown to enhance hazard susceptibility mapping, especially in areas with limited historical data (Dahri & Abida, 2017; Seejata et al., 2018; Vojtek et al., 2019).

Realtime Data Processing and Visualization. Realtime data processing and visualization capabilities are crucial for effective emergency response and decision-making during disasters. Cloud-based GIS platforms with real-time data processing capabilities significantly improve emergency response efficiency by providing decisionmakers with up-to-date situational awareness and predictive analytics (Hendriawan Nur et al., 2022; Lee et al., 2025; Yoo & Choi, 2019). The implementation of 3D visualization technologies and Web-GIS applications enables intuitive representation of complex spatial data, facilitating better understanding of hazard zones and evacuation routes among both emergency responders and affected populations (Santhanavanich et al., 2020; Shkedova et al., 2024; Yao et al., 2021).

Community-Based Approaches. Community-based approaches emerge as critical success factors, particularly in developing

country contexts. Studies consistently demonstrate that community engagement, local knowledge integration, and participatory planning significantly improve evacuation compliance and overall disaster resilience (Afolabi et al., 2023; Andrada et al., 2020; Cvetković, 2024). However, successful implementation requires sustained investment in capacity building, technical training, and stakeholder coordination (Arao, 2020; Campolet et al., 2025; Torno, 2024). Community-driven evacuation planning, supported by mobile applications and web-based platforms, has shown effectiveness in disaster-prone areas (Campolet et al., 2025; Inoguchi, 2024; Torno, 2024).

Mobile and Web-Based Applications. Mobile and web-based applications represent critical interfaces between technological systems and end users, requiring careful attention to usability, accessibility, and cultural appropriateness. Studies demonstrate that effective emergency applications must balance comprehensive functionality with intuitive design while supporting diverse user populations with varying technological literacy (Ermino et al., 2022; Lee et al., 2025; Shitole et al., 2024). Realtime visualization capabilities, particularly through 3D representations and interactive maps, enhance user understanding of hazard zones

and evacuation routes (Santhanavanich et al., 2020; Shkedova et al., 2024; Yao et al., 2021). Applications integrating hazard detection, mapping, and automated reporting capabilities demonstrate the potential of comprehensive mobile systems for disaster management (Shitole et al., 2024; de los Santos, 2024).

The Philippine Context and Implementation Challenges. The Philippine context presents specific challenges and opportunities for implementing advanced disaster management systems. While the country's high disaster vulnerability creates urgent needs for improved systems, resource constraints, infrastructure limitations, and varying technical capacities across regions require careful adaptation of technological solutions (Gacu et al., 2022; Torno, 2024). Success depends on developing accessible, cost-effective systems that leverage opensource technologies while addressing local requirements and constraints (Campolet et al., 2025; Hendriawan Nur et al., 2022). The integration of local knowledge with scientific approaches, continuous training programs, and establishment of clear communication channels between authorities and community members are essential for successful implementation (Andrada et al., 2020; Arao, 2020; Makoto et al., 2024).

Despite significant technological advances, implementation challenges persist across multiple dimensions. Infrastructure limitations, particularly in developing countries, constrain real-time data processing and communication capabilities (Hendriawan Nur et al., 2022; Torno, 2024). Technical capacity gaps, both within government agencies and communities, limit effective system utilization and maintenance (Andrada et al., 2020; Campolet et al., 2025). Resource constraints, including funding limitations and competing priorities, affect system comprehensiveness and sustainability (Arao, 2020; Gacu et al., 2022). Additionally, cultural and behavioral factors affecting technology adoption and evacuation compliance remain underexplored, particularly in developing country contexts (Campolet et al., 2025; Lee et al., 2025).

Research Gaps and Future Directions. The reviewed literature identifies several critical gaps requiring further research and development. Limited studies address long-term system sustainability, particularly regarding maintenance requirements, update protocols, and capacity building needs (Inoguchi, 2024; Torno, 2024). Few investigations examine cross-platform integration challenges when combining diverse technologies, data sources, and stakeholder systems (GonzálezVilla et al., 2022; Yoo & Choi, 2019). Furthermore,

the application of real-time visualization technologies, including 3D mapping and Web-GIS applications, for evacuation guidance warrants additional investigation (Santhanavanich et al., 2020; Shkedova et al., 2024; Yao et al., 2021).

Moving forward, successful disaster management systems must adopt integrated approaches that combine technological sophistication with practical accessibility. Cloud-based architectures provide scalable foundations while opensource technologies enable cost-effective implementation (Ahmed et al., 2020; Hendriawan Nur et al., 2022). GIS-based spatial analysis, enhanced by predictive analytics and real-time data integration, supports comprehensive risk assessment and evacuation planning (Aydin & Birincioğlu, 2022; Ghorbanzadeh et al., 2018; Parajuli et al., 2023). Realtime visualization systems provide critical monitoring and communication capabilities that enhance situational awareness and emergency response coordination (Birajdar et al., 2020; Lee et al., 2025; Santhanavanich et al., 2020). Community engagement and capacity building ensure system sustainability and effectiveness across diverse contexts (Afolabi et al., 2023; Andrada et al., 2020; Cvetković, 2024).

Alignment with SafeRoute+. The SafeRoute+ system concept aligns with these integrated approaches, proposing cloud and

GIS-based solutions specifically adapted to Philippine contexts. By incorporating predictive analytics, real-time monitoring, dynamic visualization capabilities, and community-oriented interfaces, such systems can address identified gaps while building on demonstrated best practices from international and local implementations. The integration of multiple technological components including cloud computing, GIS, predictive analytics, and mobile applications within a unified platform represents a promising direction for advancing disaster management capabilities in resource-limited settings.

CHAPTER III

METHODOLOGY

The project developers created a comprehensive plan to implement the SafeRoute+ system. This approach allows local government representatives and disaster response personnel to access the SafeRoute+ platform and obtain organized information on evacuation centers, hazard zones, and suggested evacuation routes. Through this system, the gathering and updating of disaster related information can be performed more quickly and efficiently. SafeRoute+ is hosted on a web-based platform and requires an internet connection for users to log in and utilize its features. Registered users may view available maps, shelter updates, and generated route suggestions directly through the system.

Development Method

The researchers used the Agile Method in developing the SafeRoute+ system. Agile was selected because it supports continuous adjustments and improvements during development, which is necessary for a system that integrates cloud computing, GIS mapping, and predictive analytics. The method also allows the team to review each sprint and gather feedback

that helps refine the system. This approach is suitable for SafeRoute+ since the project focuses on accuracy, responsiveness, and usability during emergency situations.

The development of SafeRoute+ followed the Agile Software Development Life Cycle, which divides the project into several phases. Each phase contributes to the completion of the system and ensures that the required features are created with flexibility and careful planning.

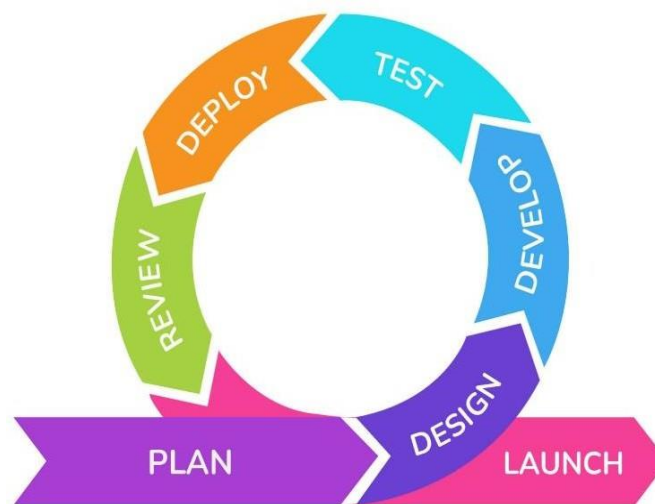


Figure 2. Agile Model

The research followed the following Agile phases:

1. Requirements Gathering. The researchers first identified the needs of local government representatives, disaster response teams, and community users. Consultations and informal interviews helped determine the common issues encountered during evacuation, such as locating safe shelters, monitoring capacity, and identifying safe routes.

These findings guided the team in selecting the main features needed for the SafeRoute+ platform.

2. Designing. During this phase, the researchers created the visual structure of the system. Initial wireframes were drafted for the dashboard, GIS maps, evacuation center information pages, and predictive outputs. The design focused on clarity and accessibility to ensure that users can easily understand and navigate the system, especially in urgent situations. System flowcharts and diagrams were also produced to guide the overall layout.

3. Developing. In this phase, programmers began building the system based on the approved design. Development was done in several sprints, allowing the team to concentrate on specific modules. Features such as cloud-based data storage, GIS mapping layers, route analysis, and congestion prediction were developed and tested incrementally. The researchers made adjustments along the way to ensure that each function worked as intended before proceeding to the next sprint.

4. Testing. The researchers conducted several rounds of testing to ensure the accuracy and functionality of each module. The testing included checking the loading of maps, verifying evacuation center information, validating prediction outputs, and reviewing user interface responsiveness. Errors found during testing were documented

and corrected. Repeated evaluations were performed to ensure the quality and stability of the system.

5. Deploying. After testing the system, the researchers deployed SafeRoute+ for initial use by selected LGU representatives. This phase allowed the team to gather real user feedback on system performance, usability, and clarity. The researchers used this feedback to make improvements and finalize the system for practical use during evacuation events.

Gantt Chart

To further present the development schedule of SafeRoute+, the researchers organized the project activities into a sprint-based Gantt chart aligned with the Agile methodology. Although the system was developed through iterative sprints, the chart was used as a visual scheduling tool to show the sequence, duration, and overlap of the major activities carried out during the study. It includes the initial planning and design stages, the development sprints for each major system module, as well as the recurring activities such as testing, review, refinement, and final deployment preparation. This presentation allowed the researchers to clearly illustrate how the project progressed from early preparation to continued iteration and completion.

Table 1. Agile Sprint Development

[illegible]

Task Name	October				November				December				January				February				March				April			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
6. Review																												
6.1 Sprint review and retrospective																												

Legend: Shaded cells indicate the scheduled period of each activity or sprint.

Table 1 presents the sprint-based Gantt chart of SafeRoute+ development. The chart shows the overall timeline of the project from October to April, using the visual structure of a Gantt chart while remaining aligned with the Agile method. The scheduled activities were grouped according to planning and requirements, design, sprint development, testing, deployment, and review. The shaded cells indicate the periods in which each activity or sprint was carried out. The chart also reflects that, beyond the initial sprint sequence, additional iterations, refinements, testing, integration, and documentation continued in the later months to further improve the system and prepare it for final deployment.

Requirements Specification

The Requirements Specification section describes the necessary conditions that the intended system must meet to perform properly. This also discusses the functionality of

the system, which includes providing the process and tasks that are required to produce a functional output. It is essential to discuss the practical requirements of the intended system for the benefit of future researchers.

Functional Requirements

Functional requirements describe the expected behavior of the system. These are the necessary tasks and processes that SafeRoute+ must perform for users, such as LGU representatives and disaster responders.

Table 2. Functional Requirements

Features	Description
GIS Mapping	The system must display evacuation centers on an interactive map. Users should be able to view hazard zones, nearby routes, and shelter details through the map interface.

Route Suggestion	The system should generate evacuation route suggestions based on distance, safety, and predicted congestion levels.
Evacuation Center Monitoring	The system must show updated information on shelter capacity, location, and
	accessibility. It should allow authorized users to view whether a center is available or nearing full capacity.
Predictive Analysis	The system must estimate evacuation center congestion based on population data and historical usage patterns.
User Authentication	The system should allow registered users to log in and securely access the dashboard and system tools.

Real Time Updates	The system must receive and display real time changes in capacity, hazard alerts, or route conditions whenever updated by authorized personnel.
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Non-Functional Requirements

Non-functional requirements describe how the system operates, focusing on performance, usability, reliability, and security.

Table 3. Non-Functional Requirements

Requirement	Description
Usability	The system should provide a user-friendly interface that LGU representatives and responders can easily navigate during emergencies.

Reliability	The system must operate consistently and must be accessible whenever needed, especially during disaster events.
Performance	The system should load maps, routes, and shelter information without noticeable delay to support quick decision-making.
Security	The system should protect user accounts and disaster-related information through secure login and encrypted data storage.
Compatibility	The system should run on standard web browsers and be accessible across different devices as long as there is internet connectivity.

Maintainability	The system structure must allow future developers to update maps, add shelters, and modify prediction modules without difficulty.
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System Overview

The system provides LGU representatives with access to essential disaster-related information. The features included in SafeRoute+ help users visualize evacuation center locations, hazard areas, and possible evacuation routes. The system integrates cloud computing, GIS technology, and predictive analytics to provide reliable and updated information during disaster situations.

The system also contains a predictive component that analyzes population data and shelter usage, helping LGUs prepare and plan ahead. SafeRoute+ is designed to support decision-making by giving users accurate, organized, and up-to-date information when planning or conducting evacuation procedures.

System User Interface

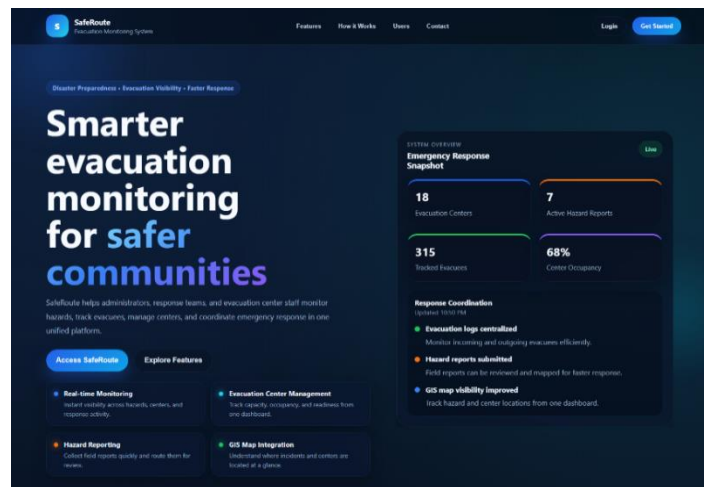


Figure 4. Home Page

The Login Page is where authorized users may access the SafeRoute+ system. Users are required to enter their registered email and password. Only verified accounts can access the system dashboard. The login module also prevents unauthorized access by ensuring secure authentication.

Hardware Specifications

The table below presents the minimum hardware requirements needed to operate SafeRoute+ during development and deployment.

Table 4. Hardware Specifications

Component	Minimum Requirement	Recommended
Processor	Intel i3	Intel i5 or higher
Memory	4 GB RAM	8GB RAM
Storage	250 GB HDD	256 GB SSD or higher
Monitor	14-15 inches	20 inches
Network	Stable connection	High-speed internet

These hardware specifications ensure that the system will function smoothly and can run all necessary processes

Software Specifications

The table below shows the required software components used in the development of SafeRoute+.

Table 5. Software Specifications

SOFTWARE	VERSION	USAGE
VS Code	Latest	Coding and development
XAMPP/Cloud DB	Latest	Backend and database

LeafletJS/MapLibre	Latest	GIS mapping engine
Node/PHP	Latest	Backend scripting
Browser	Latest	System access

These tools were chosen for their reliability, wide support, and compatibility with the system's requirements.

System Analysis and Design

System analysis and design involve understanding and specifying in detail what a system should accomplish and how the system's components should be implemented to function effectively. This process guides the developers in identifying the required features of SafeRoute+ and determining how each module should work together to support evacuation planning and disaster response.

System Overview

The proposed system, SafeRoute+: A Cloud and GIS-Based Predictive System for Evacuation Center Oversight, is a web-based platform designed to support the Provincial Disaster Risk Reduction and Management Office (PDRRMO) and its municipal counterparts. The system provides a centralized solution for organizing and managing critical evacuation-related information, including shelter capacity, hazard

locations, route accessibility, and population exposure.

SafeRoute+ can generate a unified database that stores evacuation center profiles, hazard maps, predicted congestion levels, and real-time shelter updates. By integrating cloud technology and GIS mapping, the system allows authorized personnel to view, update, and analyze essential disaster information. Predictive analytics enable the system to forecast possible overcrowding based on population estimates and historical data.

Through its features, the system supports decision-making by displaying interactive maps, visual indicators, and route suggestions. SafeRoute+ also facilitates coordination between provincial and municipal DRRM offices, providing both high-level insights and localized information. The platform aims to improve evacuation planning and ensure faster, more informed responses during emergency events.

System Architecture

This part of the research illustrates the detailed interaction between the users of the system, the system components, and the execution of tasks to ensure smooth operation.

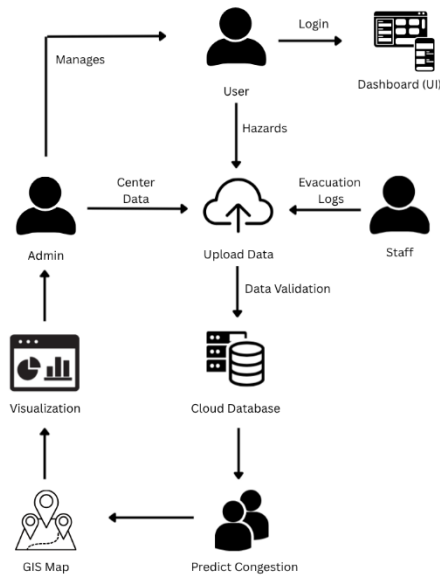


Figure 5. System Architecture

The system architecture shows how the SafeRoute+ platform handles data flow between the users, cloud database, GIS engine, and predictive module. The Super Admin, typically the PDRRMO, oversees user management, verifies updates, and manages system-wide data such as hazard layers and evacuation center profiles. Municipal DRRM officers input local updates into the system, including shelter availability, route conditions, and new evacuation center information.

Once data is uploaded, the system processes it and displays the updated results on the dashboard. GIS mapping layers render hazard maps, evacuation sites, and route

suggestions, while the predictive model analyzes congestion risk based on population exposures and historical patterns. SafeRoute+ acts as the central hub, ensuring coordinated information sharing, real-time updates, and seamless communication among all user roles.

Use Case Diagram

This section presents the graphical layout of how the system's primary and secondary users interact with the different components and features of SafeRoute+. A use case diagram visually represents how various actors perform actions within the system. It outlines the system's functionality and highlights the roles and permissions of each user group.

Use case diagrams help developers, clients, and future researchers understand the scope of the system and the relationship between actors and functions.

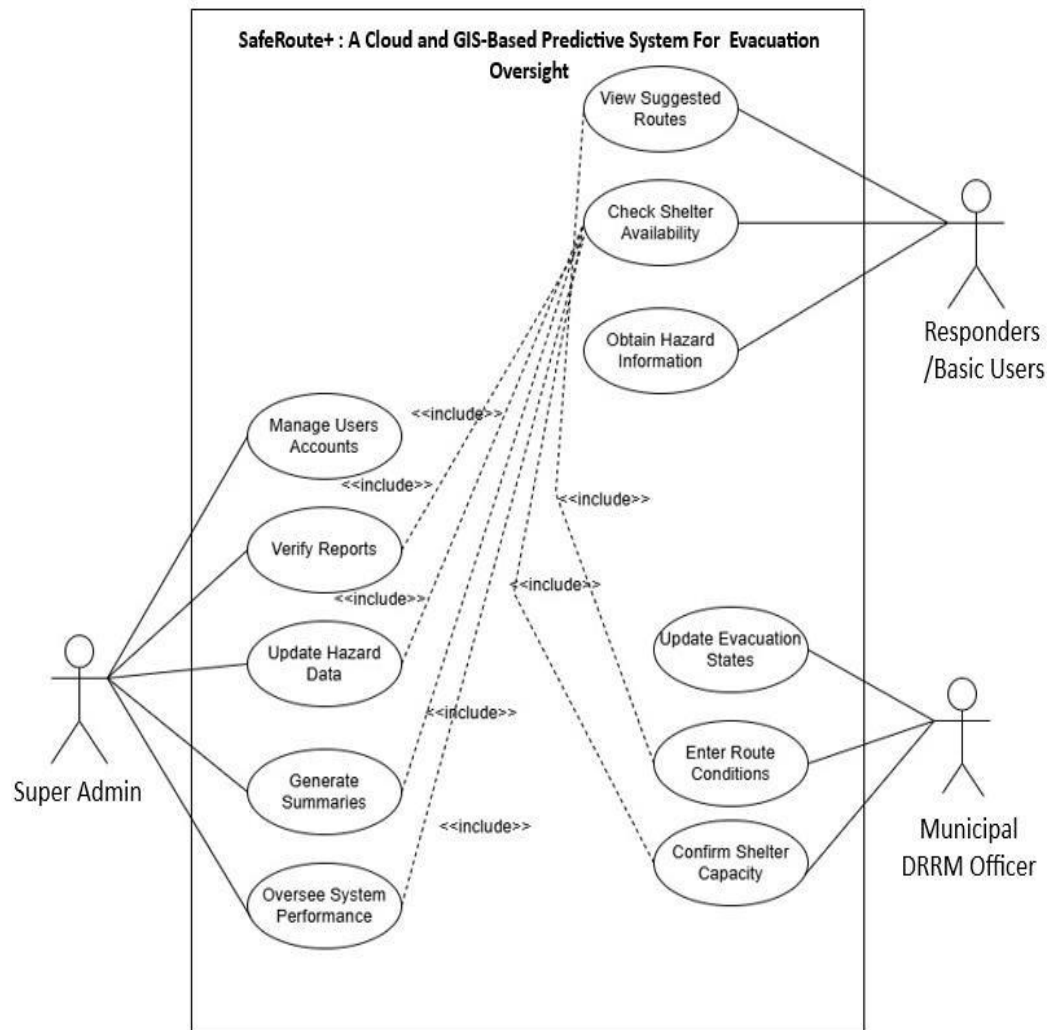


Figure 6. Use Case Diagram

In the diagram, the Super Admin has the highest level of access, including managing user accounts, verifying municipal reports, updating hazard data, generating province-wide summaries, and overseeing system performance. Municipal DRRM officers handle localized tasks such as updating evacuation center status, entering route conditions, and confirming shelter capacity reports. Responders and basic users access

the system for viewing suggested routes, checking shelter availability, and obtaining hazard information as needed.

Activity Diagram

An activity diagram represents the flow of processes within the system and shows how users interact with the features of SafeRoute+. It visualizes the movement of data, user decisions, and the transitions between different system states. This diagram helps illustrate how the system responds to user actions and how tasks are executed in sequential or parallel steps.

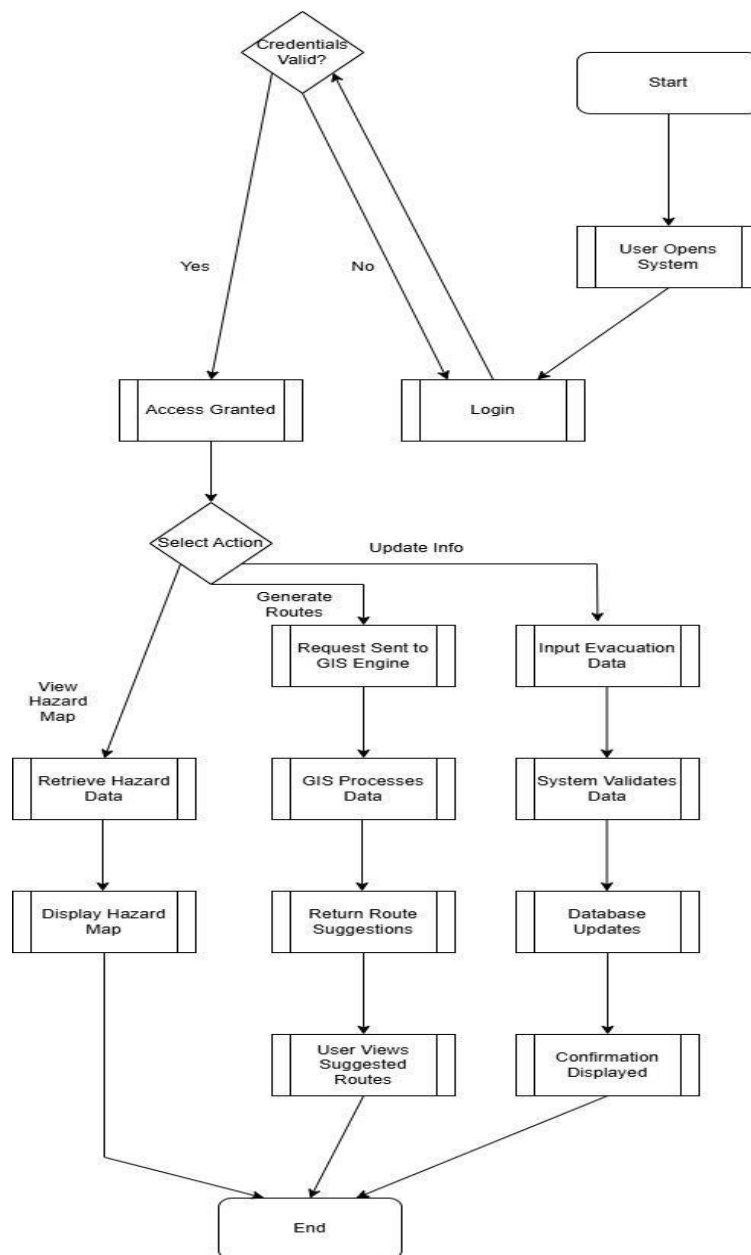


Figure 7. Activity Diagram

The diagram above explains user workflows and system operations, including how evacuation data is uploaded, processed, and displayed. It demonstrates how users—such as PDRRMO and MDRRMO personnel—perform actions like logging in,

updating evacuation center information, generating route suggestions, or viewing hazard maps. The activity diagram helps define the system's behavior and the sequence of actions necessary to complete each function.

Data Flow Diagram

This diagram provides an overview of how information moves throughout the SafeRoute+ system. It shows the main processes involved, the flow of data between these processes, and the external entities interacting with the system. The DFD helps explain how user inputs, database interactions, and system outputs are connected.

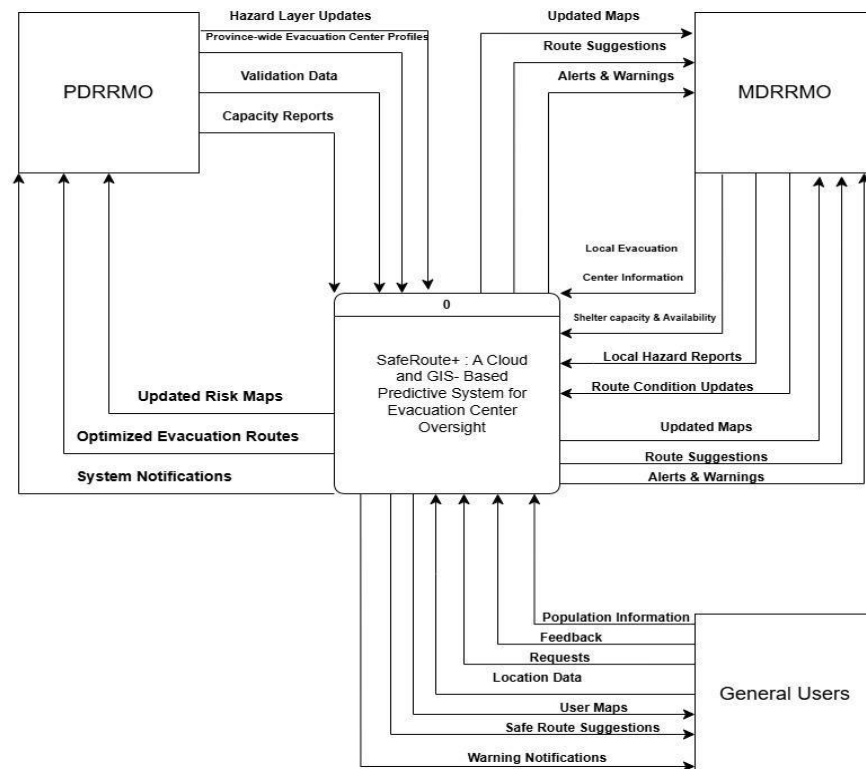


Figure 8. Context Diagram

This figure illustrates how external entities such as PDRRMO, MDRRMO, and system users interact with the central SafeRoute+ platform. It displays the movement of data inputs like shelter capacity, hazard levels, and population information, and how outputs such as maps, reports, and route suggestions return to the users.

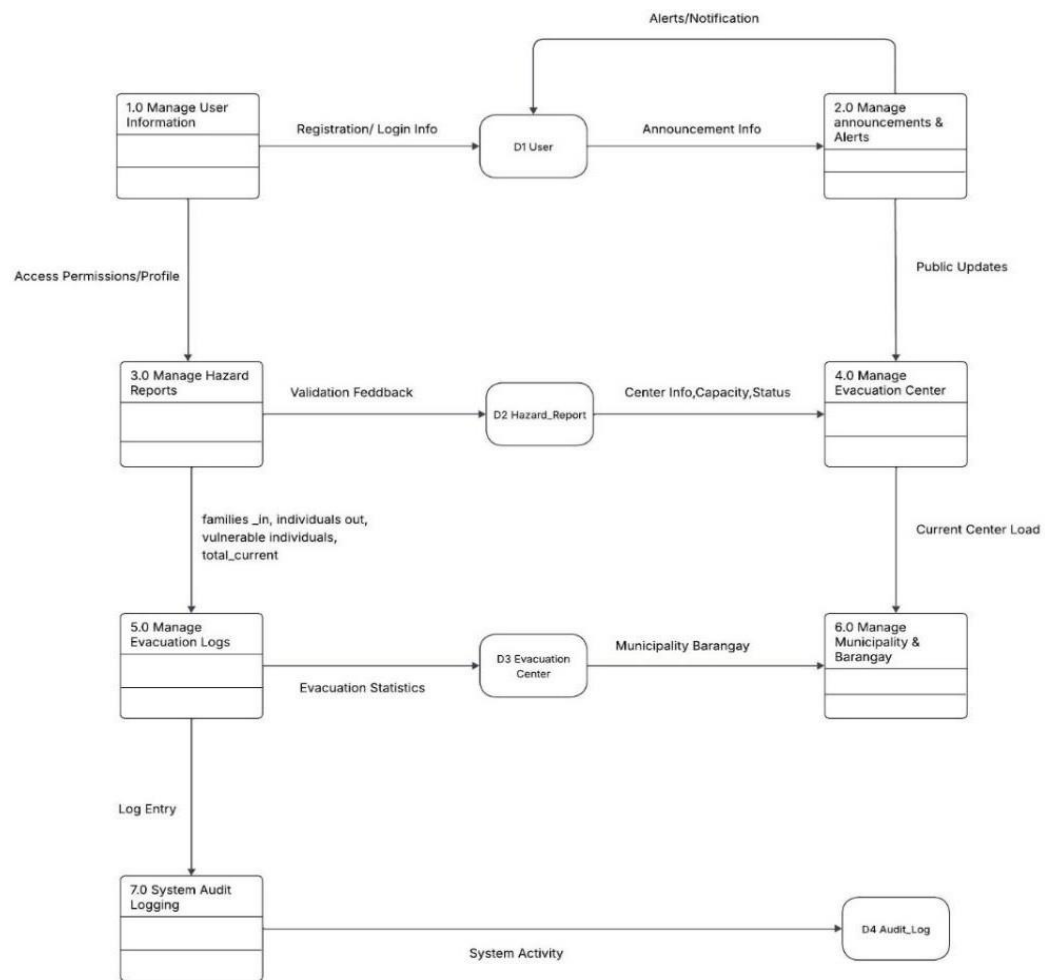


Figure 9. Data Flow Diagram (Level 1)

This diagram expands the main processes of the system, including data input, processing, route analysis, congestion

prediction, and map rendering. It also shows how data passes through the cloud database and GIS modules before being displayed to the user interface.

ERD (Entity Relationship Diagram)

The ERD displays the logical structure of the database used by SafeRoute+. It shows the relationships between tables, their attributes, and how data is connected throughout the system. This helps future developers understand how information is stored, retrieved, and updated.

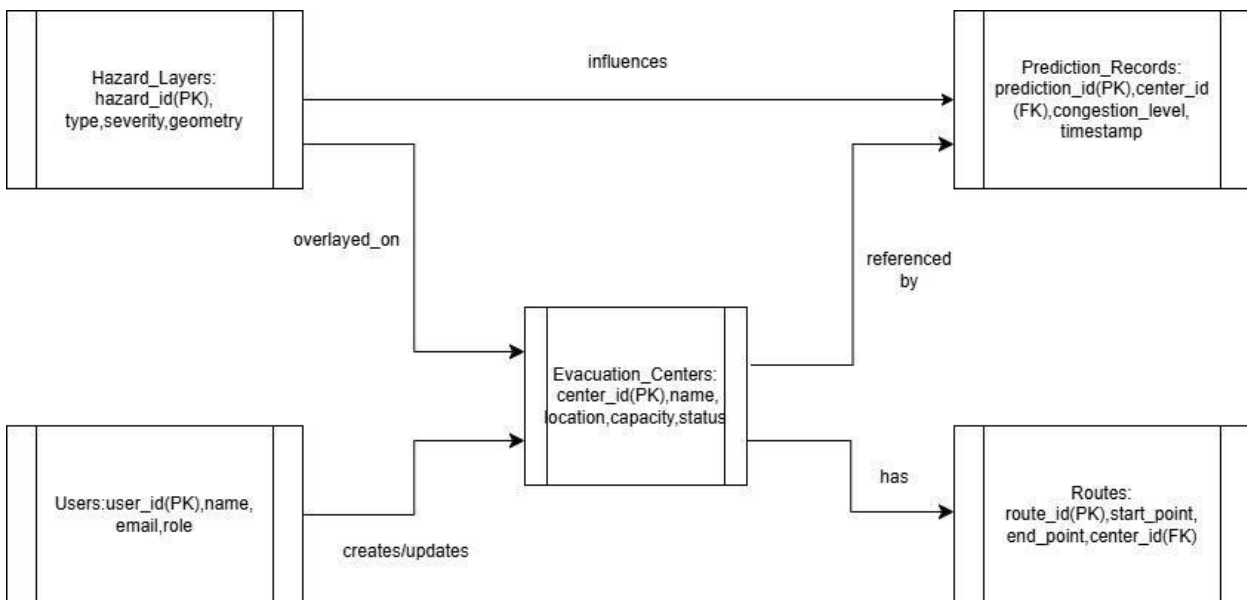


Figure 10. Entity Relationship Diagram

The ERD shows the various tables used by the system such as Users, Evacuation Centers, Hazard Layers, Routes, and

Prediction Records. It also identifies the primary keys, foreign keys, and cardinalities between these tables, illustrating how the system maintains structured and relational data

Database Schema

Below is the updated database schema for SafeRoute+. The schema was expanded to reflect all essential data components required for evacuation center management, hazard monitoring, and predictive analysis.

The schema includes user accounts, shelters, hazard zones, route data, prediction outputs, and system activity logs. This structure allows efficient storage, retrieval, and processing of spatial and non-spatial data.

evacuation center oversight using cloud technologies, GIS mapping, and predictive analytics. Since the development of the system is still ongoing, the evaluation procedures described in this section represent the planned methods for verifying the functionality and performance of the prototype once all core modules are completed.

The project developers will conduct a series of structured tests to examine system behavior under expected conditions. Functional testing will confirm that each module performs according to its intended purpose. This includes verifying GIS layer loading, evacuation center data processing, real time capacity updates, route generation, and predictive congestion outputs. The testing will also include consistency checks for spatial data and system responses during hazard simulations.

Usability testing will be carried out to determine whether the interface is intuitive and accessible for PDRRMO staff and municipal DRRM officers. Since evacuation tools must be simple and clear, usability testing will focus on navigation clarity, map readability, and system responsiveness. Predictive model validation will be performed by comparing system generated estimates with sample historical data provided by the PDRRMO.

System performance testing will assess how well the cloud-based database and mapping components respond under loading conditions. This includes checking map rendering speed, data request handling, and server response time. Errors encountered during testing will be recorded and addressed before preparing the system for deployment.

These testing procedures are designed to ensure that SafeRoute+ meets the intended reliability and usability standards required for disaster management applications.

Participants of the Study

This section identifies the individuals who will take part in the system testing and validation stage once SafeRoute+ is ready for evaluation. Since the project does not include surveys or formal user acceptability tests, the participants will function mainly as technical validators and operational reviewers.

The participants were selected based on their expertise and direct involvement in disaster risk reduction. Their insights will help ensure that the system reflects real-world evacuation workflows and provides accurate and

actionable information.

Table 6. Participants of the Study

Participant Group	Role in the Study	Purpose
Provincial DRRM Officers (PDRRMO)	System validators	Verify provincial level evacuation center data, review map accuracy, assess workflow relevance
Municipal DRRM Officers	Operational reviewers	Validate local evacuation routes, update local shelter information, check usability
Project Developers	Technical testers	Debug modules, document errors, conduct iterative functional testing

These participants will contribute to the refinement of SafeRoute+ and ensure that the system aligns with operational requirements of evacuation planning.

Likert Scale

Although the project does not currently include survey-based user acceptability testing, the Likert Scale is included as part of the standard methodology format. It may be used in future system evaluations once the platform is deployed to PDRRMO and municipal offices.

The Likert Scale is intended to measure perceptions on system usefulness, accuracy, usability, interface quality, and reliability. If used in future research, this scale will support systematic analysis of stakeholder feedback.

Table 7. Likert Scale Interpretation

Range	Interpretation
4.21 to 5.00	Strongly Agree
3.41 to 4.20	Agree
2.61 to 3.40	Neutral
1.81 to 2.60	Disagree

This scale provides a clear method for interpreting responses, should user perception measurement be needed for further enhancement of SafeRoute+.

Implementation Plan

The implementation plan outlines the expected timeline for completing the development of SafeRoute+. This includes the major activities necessary for system preparation, database setup, prototyping, development, and testing. The schedule below reflects the estimated duration required for each phase while considering available resources and coordination with the PDRRMO. These activities are arranged to ensure that the system is delivered within a manageable period and prepared for future deployment across municipal DRRM offices.

Table 8. Implementation Plan Overview

Activities	Duration	Start Date	End Date
Client Consultation & Sprint Reviews (PDRRMO)	Ongoing (monthly/bi-monthly)	Oct 10, 2025	May 4, 2026
Data Gathering & Requirements Analysis	3 weeks	Oct 10, 2025	Oct 31, 2025
Planning & System Design	3 weeks	Nov 1, 2025	Nov 21, 2025
Predictive Analytics Development (Model	4 weeks	Nov 22, 2025	Dec 20, 2025

Research & Prototyping)			
Sprint-Based Development & Iterations (Core Modules)	5 months	Dec 4, 2025	April 30, 2026
Cloud and GIS Setup	1 month	Jan 23, 2026	March 5, 2026
Integration of Predictive Analytics	3 weeks	Feb 16, 2026	Mar 10, 2026
System Testing & Debugging	2 months	Feb 18, 2026	Apr 3, 2026

This implementation timeline provides a structured guide for the development of SafeRoute+. Adjustments may be made depending on development progress and client feedback from the PDRRMO.

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